# Day 19: Odds, odds ratios, and logistic regression

> **Note**
>
> These are reference notes for Day 19. They cover the same material as the Day 19 slides and the self-paced tutorial. Whenever you run into a check-in or a question in the tutorial, write your answer down here.

## Where we're going

So far, our response variable has always been **quantitative**. Today the response is **binary** (yes/no, success/failure), so we need a new model: **logistic regression**.

## Data: fake news

150 news articles, some real and some fake. Our goal is to explain `type` using features of the article.

- `type`: "real" or "fake"; `type_num`: 0 (real) or 1 (fake)
- `title_words`: number of words in the title
- `title_has_excl`: whether the title has an exclamation point
- `title_caps_percent`: percent of the title in capital letters

A "success" ($Y = 1$) is a _____ article. Always check how your response is coded before modeling!

**Q1.** Before we do anything else, how do you think `title_has_excl` and `title_caps_percent` are related to `type_num`?

## Probability and odds

If $\pi$ is the probability of an event, the **odds** of the event are

$$\text{odds} = \frac{\pi}{1-\pi} = \frac{P(\text{event})}{P(\text{not event})} \qquad \pi = \frac{\text{odds}}{1+\text{odds}}$$

| Probability $\pi$ | Odds $\pi/(1-\pi)$ | How we'd say it |
|---|---|---|
| 0.5 | 1 | "even odds" |
| 0.75 | 3 | "3 to 1" |
| 0.2 | 0.25 | "1 to 4" |

Probabilities are between 0 and 1; odds are between 0 and $\infty$.

## Example: exclamation points

| | fake | real | Sum |
|---|---|---|---|
| FALSE | 44 | 88 | 132 |
| TRUE | 16 | 2 | 18 |
| Sum | 60 | 90 | 150 |

| | Title has exclamation | No exclamation |
|---|---|---|
| $P(\text{fake})$ | $16/18 = 0.889$ | _____ |
| $\text{Odds}(\text{fake})$ | $16/2 = 8$ | _____ |

Shortcut: from a table, odds = (number of successes) / (number of failures).

The **odds ratio** compares the odds in two groups:

$$OR = \frac{\text{Odds}(\text{fake} \mid \text{exclamation})}{\text{Odds}(\text{fake} \mid \text{no exclamation})} = \frac{8}{_____} = 16$$

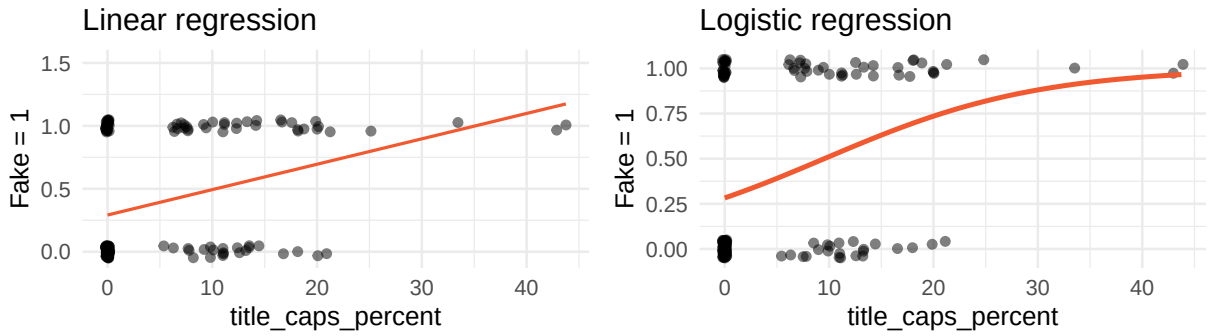
**Interpretation:** The odds of being fake are 16 times as high for articles with an exclamation point in the title as for articles without one.

> ⚠ Common mistake
>
> An odds ratio is **not** a ratio of probabilities. It is *not* correct to say that articles with an exclamation point are "16 times as likely" to be fake: the ratio of probabilities is only about 2.67.

**Q2.** An odds ratio of exactly 1 means:

An odds ratio less than 1 means:

## Why not linear regression?




A line keeps going forever, so it predicts "probabilities" outside of $[0, 1]$: the linear model predicts 1.17 for the title with the most capital letters (43.75%). Logistic regression gives an S-shaped curve that stays between 0 and 1.

## The logistic regression model

$$\log\left(\frac{\pi(x)}{1-\pi(x)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

where $\pi(x) = P(Y = 1 \mid x)$. The left side is called the **log-odds** or the **logit**. It can be any number, so it's OK for the right side to be a line.

| Scale | Formula | Range | Linear in the $x$'s? |
|---|---|---|---|
| Probability | $\pi$ | 0 to 1 | _____ |
| Odds | $\pi/(1-\pi)$ | 0 to $\infty$ | _____ |
| Log-odds | $\log(\pi/(1-\pi))$ | $-\infty$ to $\infty$ | _____ |

## Fitting the model in R

```
news_glm <- glm(type_num ~ title_caps_percent, data = news, family = binomial)
```

| term | estimate | std. error | z | p-value |
|---|---|---|---|---|
| Intercept | -0.932 | 0.220 | -4.24 | <0.001 |
| title_caps_percent | 0.098 | 0.025 | 3.84 | <0.001 |

$$\hat{\eta} = \log\left(\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)}\right) = -0.932 + 0.098x$$

where $x$ is `title_caps_percent`.

## Interpreting the slope

$\hat{\beta}_1 = 0.098$ is the change in the **log-odds** for a 1-unit increase in $x$. To get to the odds scale, exponentiate:

$$\widehat{\text{odds}}(x) = e^{\hat{\beta}_0 + \hat{\beta}_1 x} \qquad \Longrightarrow \qquad \frac{\widehat{\text{odds}}(x+1)}{\widehat{\text{odds}}(x)} = \frac{e^{\hat{\beta}_0} e^{\hat{\beta}_1 (x+1)}}{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x}} = e^{\hat{\beta}_1}$$

So $e^{\hat{\beta}_1}$ is an **odds ratio**. Here, $e^{0.098} \approx 1.103$:

> For each additional percentage point of capital letters in the title, the _____ of the article being fake _____ by about 1.10 (an increase of about _____).

For a $c$-unit increase in $x$, the odds are multiplied by $e^{c\hat{\beta}_1}$.

What happens to the odds of fake news if the percent of capital letters increases by 10 percentage points?

> The odds _____ by _____

> **Common mistakes**
>
> - $\hat{\beta}_1$ is **not** the change in the probability. In logistic regression, the change in probability for a 1-unit increase in $x$ depends on $x$.
> - $e^{\hat{\beta}_1}$ **multiplies** the odds; it is not added to them.
> - $e^{\hat{\beta}_1}$ multiplies the **odds**, not the probability.

## From log-odds to probability

What's the estimated probability that an article with **20%** capital letters in the title is fake? Work from the inside out:

1. **Log-odds:** $\hat{\eta} = -0.932 + 0.098(20) = 1.023$
2. **Odds:** $\hat{\omega} = e^{\hat{\eta}} = e^{1.023} = 2.78$
3. **Probability:** $\hat{\pi} = \dfrac{\hat{\omega}}{1+\hat{\omega}} = \dfrac{e^{\hat{\eta}}}{1+e^{\hat{\eta}}} = \dfrac{2.78}{1+2.78} = 0.735$

The intercept works the same way: at $x = 0$, the log-odds are _____, the odds are _____, and the probability is _____.

**In R**, `predict()` does this for you, but by default it gives the **log-odds**. Use `type = "response"` to get probabilities:

```
predict(news_glm, newdata = data.frame(title_caps_percent = 20))
```

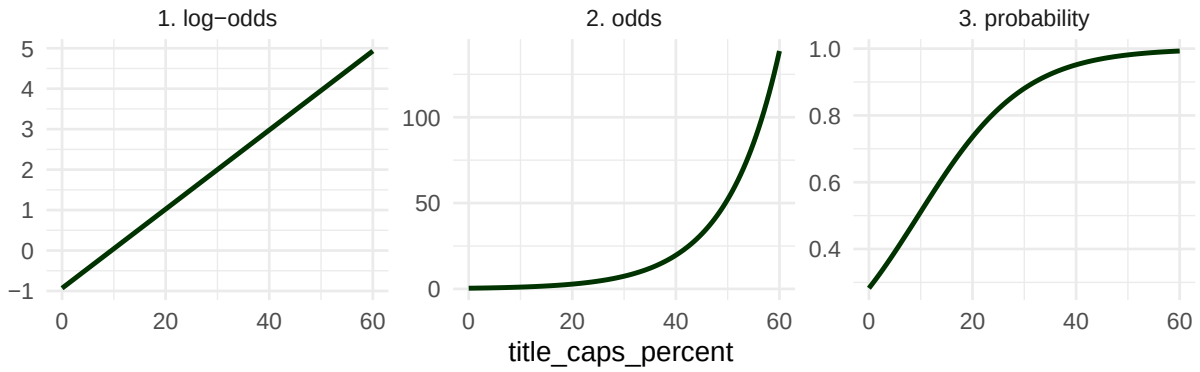
```
       1
1.022656
```

```
predict(news_glm, newdata = data.frame(title_caps_percent = 20), type = "response")
```

```
        1
0.7354896
```

*Note:* The first line of output (without type = "response") is on which scale? _____

## The fitted model on three scales




**Q3.** Describe the shape of the fitted model on each of the three scales. Why does the probability curve flatten out near 1?

## Moving between the three scales

| From | To | Formula | In R |
|---|---|---|---|
| log-odds $\eta$ | odds | $e^{\eta}$ | `exp(eta)` |
| odds | probability | $\frac{\text{odds}}{1+\text{odds}}$ | `odds / (1 + odds)` |
| log-odds $\eta$ | probability | $\frac{e^{\eta}}{1+e^{\eta}}$ | `exp(eta) / (1 + exp(eta))` |
| probability $\pi$ | odds | $\frac{\pi}{1-\pi}$ | `p / (1 - p)` |
| odds | log-odds | $\log(\text{odds})$ | `log(odds)` |

## A binary predictor gives back the odds ratio

```
excl_glm <- glm(type_num ~ title_has_excl, data = news, family = binomial)
exp(coef(excl_glm))
```

```
      (Intercept) title_has_exclTRUE
              0.5               16.0
```

- $e^{\widehat{\beta}_1} = 16$ is exactly the odds ratio we computed from the table.

**Q4.** What does $e^{\widehat{\beta}_0} = 0.5$ represent in context?

## Your turn: three predictors

`glm(type_num ~ title_has_excl + title_caps_percent + title_words, data = news, family = binomial)`.

**Q5.** Write out the fitted model on the log-odds scale for articles with and without an exclamation point in the title.

(A) With an exclamation point:

$$\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = _____ + _____ \times \texttt{title_caps_percent} + _____ \times \texttt{title_words}$$

(B) Without an exclamation point:

$$\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = _____ + _____ \times \texttt{title_caps_percent} + _____ \times \texttt{title_words}$$

**Q6.** Interpret each slope coefficient in terms of the **odds** of being fake news. (*Note:* templates on last page if you get stuck; but try it on your own first)

`title_has_excl`:

`title_caps_percent`:

`title_words`:

**Q7.** The estimated probability that an article with an exclamation point, 10% capital letters, and 12 words in the title is fake: _____

**Q8.** Which predictors appear to be most important for this model? Why?

**Q9.** What questions do you have about the `summary()` output?

## Interpretation templates

| Quantity | Interpretation |
|---|---|
| $\widehat{\beta}_j$ | Change in the log-odds of success for a 1-unit increase in $x_j$, holding other predictors constant |
| $e^{\widehat{\beta}_j}$, quantitative $x_j$ | For a 1-unit increase in $x_j$, the odds of success are multiplied by $e^{\widehat{\beta}_j}$, holding other predictors constant |
| $e^{\widehat{\beta}_j}$, indicator $x_j$ | The odds of success are $e^{\widehat{\beta}_j}$ times as high for the indicated group as for the reference group, holding other predictors constant |
| $e^{\widehat{\beta}_0}$ | The odds of success when all predictors equal 0 (or for the reference group) |

## R function reference

| Function | Purpose |
|---|---|
| `table(x, y) \|> addmargins()` | Two-way table with row and column totals |
| `glm(y ~ x, data, family = binomial)` | Fit a binary logistic regression model |
| `coef(model)`, `exp(coef(model))` | Coefficients on the log-odds scale; odds ratios |
| `summary(model)` | Model summary and Wald tests |
| `predict(model, newdata)` | Predictions on the log-odds scale |
| `predict(model, newdata, type = "response")` | Predictions on the probability scale |